

Human exposure to PBDEs: associations of PBDE body burdens with food consumption and house dust concentrations.



This study was designed to determine the body burden of polybrominated diphenyl ethers (PBDEs) among first-time mothers in the Greater Boston, Massachusetts area and to explore key routes of exposure. We collected breast milk samples from 46 first-time mothers, 2-8 weeks after birth. We also sampled house dust from the homes of a subset of participants by vacuuming commonly used areas. Data on personal characteristics, diet, home furniture, and electrical devices were gathered from each participant using a questionnaire. Breast milk and dust samples were analyzed for PBDEs using gas chromatography/ mass spectrometry. PBDE concentrations were log-normally distributed in breast milk and dust. We found statistically significant, positive associations between PBDE concentrations in breast milk and house dust ($r = 0.76$, $p = 0.003$, not including BDE-209), as well as with reported dietary habits, particularly the consumption of dairy products ($r = 0.41$, $p = 0.005$) and meat ($r = 0.37$, $p = 0.01$). Due to low detection rates, it was not possible to draw conclusions about the association between BDE-209 in milk and dust. Our results support the hypothesis that the indoor environment and diet both play prominent roles in adult human exposure to PBDEs.